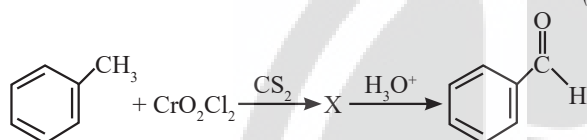


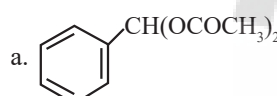
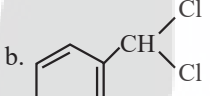
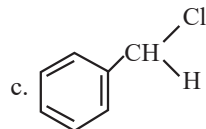
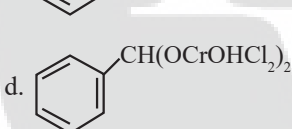
# CHAPTER 9

## Aldehydes, Ketones and Carboxylic Acids

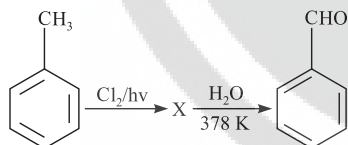
### Preparation of Aldehydes and Ketones

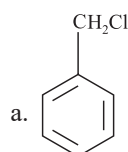
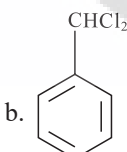
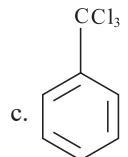
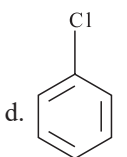
1. The intermediate compound 'X' in the following chemical reaction is: (2021)



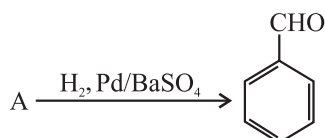
- a.  b.   
c.  d. 

2. Identify compound X in the following sequence of reactions: (2020)



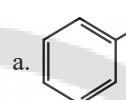
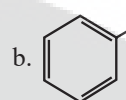
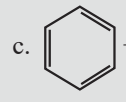
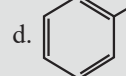
- a.  b.   
c.  d. 

3. Identify compound (A) in the following reaction: (2020-Covid)



- a. Toluene b. Acetophenone  
c. Benzoic acid d. Benzoyl chloride

4. Reaction by which Benzaldehyde cannot be prepared: (2013)

- a.  +  $\text{CrO}_2\text{Cl}_2$  in  $\text{CS}_2$  followed by  $\text{H}_3\text{O}^+$   
b.  +  $\text{H}_2$  in presence of  $\text{Pd-BaSO}_4$   
c.  +  $\text{CO} + \text{HCl}$  in presence of anhydrous  $\text{AlCl}_3$   
d.  +  $\text{Zn/Hg}$  and concentration  $\text{HCl}$

### Physical Properties and Chemical Reactions of Aldehydes and Ketones

5. Given below are two statements: (2022)

Statements: I The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular mass because of weak molecular association in aldehydes and ketones due to dipole - dipole interactions.

Statements: II : The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular mass due to the absence of H-bonding.

In the light of the above statements, choose the most appropriate answer from the options given below

- a. Statements I is incorrect but Statements II is correct.  
b. Both Statements I and Statement II are correct.  
c. Both Statements I and Statements II are incorrect.  
d. Statements I is correct but Statements II is incorrect.

6. Match List-I with List-II. (2022)

#### List-I

(Products formed)

- (a) Cyanohydrin  
(b) Acetal  
(c) Schiff's base  
(d) Oxime

#### List-II

(Reaction of carbonyl compound with)

- (i)  $\text{NH}_2\text{OH}$   
(ii)  $\text{RNH}_2$   
(iii) alcohol  
(iv)  $\text{HCN}$

d. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

CC(C)C(=C)C(=O)C

#### d. Aldol condensation

d. 

Identify A, X, Y and Z

d

### c. Ketone

d. Carboxylic acid

c. Acetone

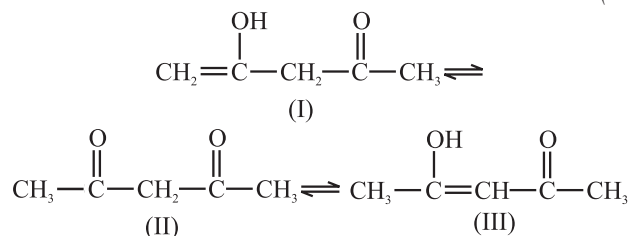
#### d. Ozone

d. A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as aldehyde-ketone equilibration

d. Hydrocyanic acid

d. Pentanal

17. The order of stability of the following tautomeric compounds is: (2013)



- a. II > I > III      b. II > III > I  
c. I > II > III      d. III > II > I

### Methods of Preparation of Carboxylic Acids

18.  $\text{RMgX} + \text{CO}_2 \xrightarrow[\text{ether}]{\text{dry}} \text{Y} \xrightarrow{\text{H}_3\text{O}^+} \text{RCOOH}$  (2022)

What is Y in the above reaction?

- a.  $(\text{RCOO})_2\text{Mg}$       b.  $\text{RCOO-Mg}^+\text{X}$   
c.  $\text{R}_3\text{CO-Mg}^+\text{X}$       d.  $\text{RCOO-X}^+$

### Physical Properties and Chemical Reactions of Carboxylic Acids

19.  $\text{CH}_3\text{CH}_2\text{COO}^-\text{Na}^+ \xrightarrow[\text{Heat}]{\text{NaOH, ?}} \text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$ .

Consider the above reaction and identify the missing reagent/chemical. (2021)

- a. Red Phosphorus      b. CaO  
c. DIBAL-H      d.  $\text{B}_2\text{H}_6$

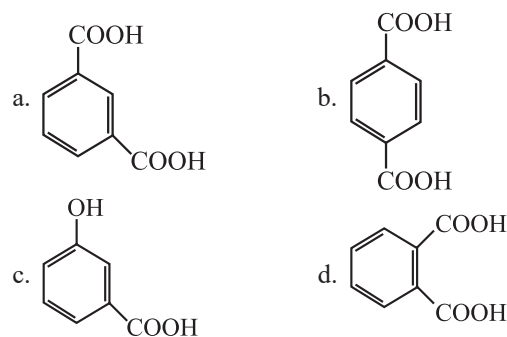
20. Match List-I with List-II. (2021)

| List-I |                                                                                                                                                       | List-II |                                |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------|
| (A)    |  $\xrightarrow[\text{AlCl}_3/\text{CuCl}]{\text{CO, HCl, Anhyd.}}$ | (i)     | Hell-Volhard-Zelinsky reaction |
| (B)    | $\text{R}-\text{C}(=\text{O})-\text{CH}_3 + \text{NaOX} \rightarrow$                                                                                  | (ii)    | Gattermann-Koch reaction       |
| (C)    | $\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$                                                     | (iii)   | Haloform reaction              |
| (D)    | $\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$                                                  | (iv)    | Esterification                 |

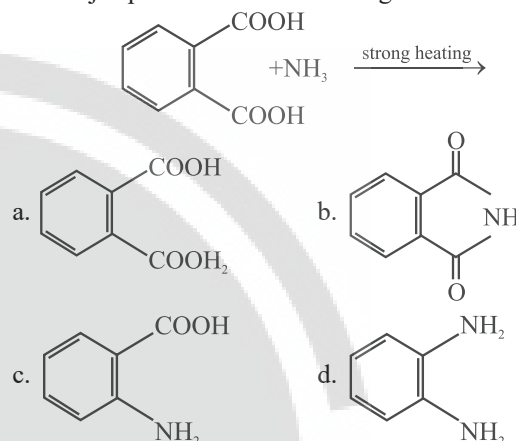
Choose the correct answer from the options given below.

- a. A-iii B-ii C-i D-iv  
b. A-i B-iv C-iii D-ii  
c. A-ii B-iii C-iv D-i  
d. A-iv B-i C-ii D-iii

21. Which of the following acid will form an (i) Anhydride on heating and (ii) Acid imide on strong heating with ammonia? (2020-Covid)



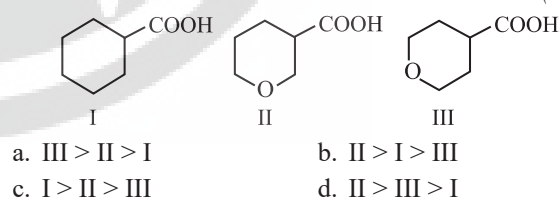
22. The major product of the following reaction is: (2019)



23. Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their: (2018)

- a. Formation of intramolecular H-bonding  
b. Formation of carboxylate ion  
c. Formation of intermolecular H-bonding  
d. More extensive association of carboxylic acid via van der Waals force of attraction

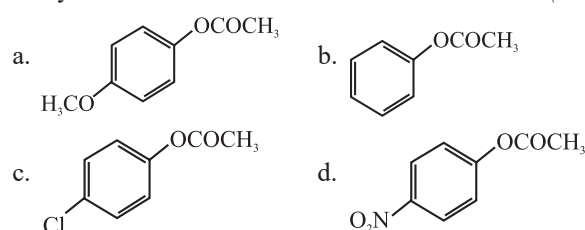
24. The correct order of strengths of the carboxylic Acids is: (2016 - II)



25. The oxidation of benzene by  $\text{V}_2\text{O}_5$  in the presence of air produces: (2015 Re)

- a. Maleic anhydride      b. Benzoic acid  
c. Benzaldehyde      d. Benzoic anhydride

26. Which one of the following esters gets hydrolysed most easily under alkaline conditions? (2015 Re)



**Answer Key**

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| d  | b  | d  | d  | b  | a  | c  | c  | c  | d  | b  | b  | c  | b  | c  | b  | d  |
| 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |    |    |    |    |    |    |    |    |
| b  | b  | c  | d  | b  | c  | d  | a  | d  |    |    |    |    |    |    |    |    |

